The Art of Uncertain Topoplots: Designing and Evaluating Uncertainty Visualizations for EEG/ERP Data

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Abstract—TODO: Supplementary-material overview

Index terms—TODO: Supplementary Material

1 Supplementary Material

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ID	Interview question
Warm-up	
Q1	Could you briefly describe your background in EEG/ERP research?
Q2	What types of EEG visualizations do you most commonly use in your work, and for what purposes?
Thinking About Uncertainty	
Q3	When you think about quantifying uncertainty in EEG or ERP data, what kinds of methods come to mind?
Q4	Do you currently include uncertainty information when you visualize EEG/ERP data? If so, what type of uncertainty do you include, and why?
Use of Topoplots	
Q5	At what stages of your research do you use topoplots, and for which audiences?
Q6	When you create a topoplot, what purposes does it serve?
Q7	Why do you think other researchers use topoplots?
Uncertainty in Topoplots	
Q8	Do you ever show uncertainty in topoplots specifically? If not, why not?
Q9	If you wanted to show uncertainty in a topoplot, how might you visualize it?
Value and Barriers	
Q10	Do you think visualizing uncertainty in topoplots would be helpful? Why or why not?
Q11	Why do you think many researchers currently do not include uncertainty in their topoplots?
Future Needs	
Q12	Is there anything you wish topoplot tools did better?
Q13	What requirements would an ideal topoplot need to meet?
Closing	
Q14	Is there anything I did not ask about that you think is important regarding topoplots or uncertainty visualization?

Table S1: Semi-structured interview guide used in nine qualitative interviews.

Paper	Field	Data	Visualizations compared	Type	Sample size	How it was measured
Wainer & Francolini [1]	Geospatial mapping	R	Univariate and bivariate colour maps	PT	N: 16	Elementary map-reading error rate; response time
Sanyal et al. [2]	General	S	Error bars, glyph size, glyph colour, surface colour	PT	N: 27	Search and counting accuracy; response time
MacEachren et al. [3]	Geospatial mapping	S	Size, colour, transparency vs. recognizable pictorial symbols	PT, SR	E1: 72; E2: 30; pilot: 31	Intuitiveness judgments; comparison accuracy; response time
Correll & Gleicher [4]	General	S	Error bars, box plots, gradients, violin plots	PJ	E1: 96; E2: 48; E3: 96	Prediction accuracy; likelihood judgments; judgment confidence
Scholz & Lu [5]	Geospatial mapping	R	Fuzziness, lightness, transparency, shape	DM, SR	N: 60	Decision choices; preference ratings and rankings
Greis et al. [6]	Weather forecasting	M	Textual estimates, SD and quantiles; line/area charts, boxplots, bar charts, shaded bars, PDF/CDF plots	DM, SR	Survey: 90; game: 44 (38 rated)	Subjective ratings; farming-game decision choices
Hullman et al. [7]	Chemical concentrations	S	HOPs, error bars, violin plots	PJ	N: 288	Probability-estimation error; total completion time
Gschwandtner et al. [8]	Event time intervals	S	Gradients, violin plots, probability plots, error bars, ambiguation	PT, PJ	N: 72	Accuracy; estimation error; response time
Correll et al. [9]	Spatial mapping	S	Adjacent maps, bivariate maps, VSUPs	PT, DM	E1: 24; E2: 24	Identification accuracy; response time; decision choices
Fernandes et al. [10]	Public transport	S	Dotplots, CDFs, PDFs, intervals, text, baseline	DM	N: 408	Expected reward relative to optimal; variability across trials
Kale et al. [11]	Employment trends	S	HOPs, error bars, line ensembles	PT	E1: 50; E2: 150	Trend-identification accuracy; perceptual sensitivity (JND); judgment confidence
Kusumastuti et al. [12]	General	S	Ensemble, summary, distribution displays	DM	E1: 118; E2: 56	Decision accuracy; response time
Sterzik et al. [13]	Biomolecular structures	M	Colour, clouds, ensembles, hulls, sausages, contours, texture, waves, noise	PT, QF	Total: 1,756; experts: 6	Mapping-direction judgments; uncertainty-estimation accuracy; qualitative expert feedback

Table S2: Empirical evaluations of uncertainty visualization techniques.
Data: S = synthetic or simulated data; R = real observational data; M = mixed real and simulated data;
Task type: PT = perceptual tasks; PJ = probability judgments; DM = decision-making tasks; SR = subjective ratings; QF = qualitative feedback.
N = number of participants; E1–E3 = separate experiments.

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